

Hurdles in the widespread usage of Big Data and IoT

Standards: For efficient working of the IoT, there should be a predefined framework followed by devices and applications to exchange data safely over wireless or wired networks. OneM2M is an organisation which publishes the preferred standards set by the major technology giants. Sources in the firm insist that there should be interoperability among varied industries such that a common platform connects smart meters, cars, watches, pacemakers, etc.

Security and privacy: Security in some sensitive applications like a biological sensor recording the human vital signs should be protected against breach of privacy. National infrastructure related data is critical for the country's security and should have appropriate safeguards against hackers. Smart home lock systems, industrial security sensors, etc, all need protection against malicious users who can trespass illegally. IoT is advantageous since it can be operated over the Internet, but it is very risky on account of that very reason. The Internet can be breached by intruders and the devices can be wrongfully used.

Network and data centre infrastructure: Data centres and infrastructure will be under duress due to the incoming data deluge. The flows can be in bursts or continuous, primarily between the applications and sensors.

Analytics tools: IoT management is complicated and to build analytics for insights is no easy task. Various platforms use different languages and professionals have to be trained to deal with each of them.

Skills: IoT and Big Data are multi-disciplinary, and professionals require a working knowledge of both fields. As the topics are relatively new, old-school technological workers need to be trained and acquainted with the new technologies. Business analysts are required to frame the questions that will best extract the data and present the outcomes to the clients. Data scientists are required to use analytical tools to derive insights and do the technical work.

Big Data and IoT complement each other to bring out the advantages of each. The technological world has realised their significance and the Big Data + IoT industry is set to become a multi-billion dollar business, with researchers and IT firms starting to realise the potential behind the hype. This alliance is the future of technology and will fundamentally change the world around us. 

References

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